Measurement Uncertainty

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Motivation

Part of a professional approach to engineering is understanding and quantifying uncertainty in your measurements. In contrast to a cursory verification ("Looks good enough to me!"), a professional design has a set of requirements and a way to verify compliance with those requirements: "Yes, the unit under test will support the required loads, here are the measurements that support that claim and this is how we know those measurements are sufficient." The last part of that statement requires an understanding of the measurement uncertainty: how do you know if your measurements are adequate?

In a scientific experiment, measurement uncertainty is critical for determining whether an experiment agrees with a theoretical prediction or not. In the past, you may have used very lenient criteria for this ("well, it's within 10% of the prediction, so that agrees pretty well"). In this course, we will attempt to use more quantifiable, professional criteria, but without diving too deeply into rigorous statistical analysis.

Controlling standard

Like other professional standards, there is a published set of definitions, procedures and guidelines for how to deal with measurement uncertainty. The BIPM (Bureau International des Poids et Mesures) "Guide to the expression of uncertainty in measurement" (GUM)[2] is a widely-accepted standard. It is available free online at `bipm.org`, or on the course D2L site. GUM is the standard for many organizations including the International Organization for Standardization (ISO) and (by extension) the many ISO-certified engineering companies.

The 134-page GUM standard is a bit of a hard read. This document attempts to summarize and highlight its core concepts. In the event of any discrepancies, the GUM standard should be considered the authoritative document.

Factors contributing to uncertainty

Uncertainty characterizes the dispersion of values that could reasonably be attributed to the measured quantity. The term "**standard uncertainty**" should be used to refer to uncertainties calculated according to the GUM guidelines.

Uncertainty in a measurement is defined in the GUM document as *a parameter that characterizes the dispersion of values that could reasonably be attributed to the measured quantity*. It is a quantification of the doubt about the measurement result. Uncertainty is distinct from *error*, which is the (generally unknown) difference between the measured value and the 'true value' of the measured quantity. Many factors may contribute to the uncertainty, including those arising from imperfect knowledge of systematic effects. The term "**standard uncertainty**" should be used to refer to uncertainties calculated according to the GUM guidelines.

Measurement uncertainty is an essential part of every measurement. Only a very few measurements can be taken exactly, usually by counting an integer number of things. Although you may be able to exactly count the number of fingers on your hand, there is a limit to how accurately you can, say, measure your height.

Uncertainty creeps in from many directions (see GUM 3.3.2). First, there are obvious limits to any measuring device.

- Your measuring device has a limited **resolution**: if you have a ruler marked in millimeters, you can't possibly measure a length to the

nearest micrometer.

- Your measuring device has a **calibration uncertainty**: how close is one millimeter on the ruler to an exact millimeter (i.e. the distance travelled by light in vacuum in $1/299792458000^{\text{th}}$ of a second)? Are there other factors ("influence quantities") such as temperature that might affect that uncertainty?

Student lab reports sometimes contain the term "human error" as a source of catch-all uncertainty. This is poor practice and results in a dishonest uncertainty estimate. Often, the term is synonymous with "mistake" – a wrong calculation, a misunderstood instruction, a misread measurement. If you made a mistake during a measurement, you should discard it and repeat the measurement. You should never artificially increase your uncertainty with a "human error" component.

But even if your measuring device is arbitrarily precise and well-calibrated, there are still inherent errors that cannot be avoided:

- Errors often arise due to **incomplete definition** of the quantity being measured, or due to **imperfect realization** of that definition. For example, your height varies considerably based on your posture. Similarly, exactly which points on the feet and head are being considered will also affect the measurement of your height. If those factors aren't part of the definition of your "height", or if they aren't perfectly controlled during the measurement (which would be impossible), then they become a source of uncertainty.
- There may be **systematic effects** on the measurement that can only be corrected to a limited perfection. For example, you might need to correct the measurement of your height for the thickness of your socks, which might not be known with high accuracy. Systematic effects should always be corrected to the best of the experimenter's ability during the analysis. Nonetheless, there will always be a limit to how well you know the systematic effects, and on how well you can correct the results. This contributes an uncertainty due to incomplete knowledge of the required value of the correction.

You should always correct measurements for known systematic effects. The measurement uncertainty reflects your uncertainty in applying that correction, not the magnitude of the correction. Explicitly apply the correction during the analysis, not when writing down the raw data. Always explain the correction in your report.

Type A uncertainty

Repeated measurements under apparently identical conditions often give variable results. There are limits on both **repeatability** (under exactly the same conditions in a short time) and **reproducibility** (under slightly different conditions or over a long time). The variance in repeated measurements gives a rigorous way to quantify many types of uncertainty.

The GUM categorizes uncertainty estimates into 2 categories, depending on how the uncertainty estimate is derived. **Type A** uncertainties are derived solely through repeated measurements and statistical methods, while all other methods of estimating uncertainty give **Type B** uncertainties.

Type A uncertainty is derived through repeated measurements using the equations in this section.

As an example of **Type A** uncertainty, perhaps you measure your height n times in a row. The best estimate for your height h is the

arithmetic mean (average) of the n values:

$$\bar{h} = \frac{1}{n}\sum_{k=1}^{n} h_k \tag{1}$$

where h_k are the measured heights. The best estimate for the standard uncertainty *of the mean* $\bar{h}$ is

$$u(\bar{h}) = \frac{s(h_k)}{\sqrt{n}} \tag{2}$$

where $s(h_k)$ is the (sample) standard deviation (implemented in Excel as STDEV.S()):

$$s(h_k) = \sqrt{\frac{1}{n-1}\sum_{j=1}^{n}(h_j - \bar{h})^2}. \tag{3}$$

Note that type A uncertainty decreases with a larger number of measurements. In the limit of an infinite number of measurements, the standard uncertainty of the mean $\bar{h}$ goes to zero.

Any uncertainty estimate that is not derived through repeated measurements is Type B; see page 5.

Type B uncertainty is any other uncertainty not derived through repeated measurements. Sometimes Type B uncertainties are used because repeated measurements would be impractical, but some sources of uncertainty simply cannot be estimated through repeated measurements. For example, calibration uncertainty is a Type B uncertainty that cannot be revealed through repeated measurements without access to a reference standard. Similarly, imperfect correction for systematic effects typically results in Type B uncertainty. Your experiment will almost always have some Type B uncertainties. Type B uncertainties will be discussed in more detail on page 5.

Quantifying "standard uncertainty" and coverage factors

If you were to average together a large number of independent measurements, each of which was subject to random errors (following equation 1), and then repeat that process many times to get a sampling of *averages* $\bar{h}_j$, then the probability distribution of those averages would follow a Gaussian (or "normal") distribution. The normal distribution would be centered at the true value, with a standard deviation given by equation 2. Amazingly, this holds true regardless of the underlying probability distribution of the errors (the "Central Limit Theorem"). The one caveat is that the number of measurements must be large: theoretically an infinite number, although some tens usually works in practice.

The central limit theorem allows rigorous definitions for many Type A uncertainties.

What that means is that we can strictly quantify Type A uncertainty. For Type A uncertainty calculated using equation 3, any given measurement will be within 1 standard deviation of the true value about

68% of the time, within 2 standard deviations about 95.4% of the time and within 3 standard deviations about 99.7% of the time. Standard measurement uncertainty can therefore best be thought of as follows:

One can expect the measurement error (i.e. the unknown difference between the measurement and the 'true' value) **to be less than the standard uncertainty about 68% of the time**.

Uncertainties are not safety factors, nor are they confidence intervals. When estimated correctly, there is a $\sim$ 32% chance that the error is larger than the standard uncertainty. If you desire a higher level of confidence, multiply the standard uncertainty by a coverage factor (usually 2 or 3) and state that clearly.

For Type A uncertainties, this follows rigorously from the Central Limit Theorem. However, this is also the best way to think of Type B uncertainties (see below).

While GUM standardizes the term "standard uncertainty" at the 68% level, there are plenty of times when you might want to discuss the uncertainty with a higher level of confidence. Nobody wants to cross a bridge they are 68% sure won't collapse. By multiplying the standard uncertainty by a **coverage factor** k, an "expanded uncertainty" can be calculated with a higher level of confidence. A coverage factor of 2 or 3 is very typical, giving a level of confidence of 95% or 99.7% respectively. For major ground-breaking discoveries such as the discovery of the Higgs boson, coverage factors as large as k=5 (99.99994% confidence) may be used.

Type B uncertainty

Any uncertainty not derived by repeated measurements and the application of equations 1 and 2 is a Type B measurement. Most uncertainties are Type B, simply because Type A uncertainties are often impractical. For example, to estimate the calibration uncertainty of an instrument using Type A methods would require purchasing some tens of the instruments, each with a separate and independent calibration against a standard. If you would prefer to simply use the manufacturer's quoted uncertainty, that's a Type B uncertainty.

Type B uncertainties are honest estimates of the uncertainty. They are usually not rigorously defined, but should express the range within which the experimenter is about 68% sure that the measurement falls.

Type B uncertainties should be honestly estimated to conform to the same definition of "uncertainty" as for Type A. This ensures that all uncertainties can be compared on a standard basis, regardless of Type. Rigorously, this requires estimating the probability distribution and then assessing its standard deviation. Approximately speaking, the Type B uncertainty should be estimated so that you're about 68% certain that the true value is within 1 uncertainty of the measurement.

Most uncertainties you will encounter are Type B.

That means that when you are evaluating a Type B uncertainty, you should not think "what range am I absolutely sure that the measurement falls in?", but rather **"what range am I about 68% sure that the measurement falls in?"** Obviously, this is not something you can typically estimate to high accuracy, but an honest experiment requires an honest effort to evaluate the Type B uncertainties.

There is no rigorous, absolute recipe for how to estimate Type B uncertainties.

There is no rigorous recipe for how to estimate Type B uncertainties. That's simply the price of using Type B uncertainty; if you want a strictly rigorous procedure then you must repeat the measurement many independent times and use Type A. For Type B, you are expected to use your best judgment to arrive at an honest, defensible estimate. Many students find this difficult; below are a few examples of how you might approach various Type B uncertainties. Keep in mind that there is often no one "right way".

EXAMPLE: Digital quantization. With a digital instrument, there's always an uncertainty in the last digit. For example, if a balance reads 10.1 g, you have no way of knowing whether the true mass is closer to 10.15 g or to 10.05 g. Assuming that every value in the range $[x - \Delta, x + \Delta]$ is equally likely (a rectangular probability distribution), it can be shown rigorously that the resulting standard deviation and hence the uncertainty is $u = \Delta/\sqrt{3} \approx 0.6\Delta$ (see GUM 4.3.7). More approximately, 68 % of the probability is contained in the range $x \pm 0.68\Delta$, giving $u \approx 0.68\Delta$. In the example with the balance, $\Delta = 0.05$ g, so the measured mass would be reported as: 10.10 g with a standard uncertainty of 0.03 g.

EXAMPLE: Rulers and analog meters. This is a case when there's no one "right way"; you must ask yourself honestly how uncertain you are of the measurement. It will depend on how small the divisions on the ruler are, how good your eyesight is, and how closely you can align the object with the ruler. Let's say you use a millimeter-scale ruler to measure a well-aligned object, and you feel "pretty confident" (about 68% confident) that the object is between 10.0 mm and 10.5 mm. It would be reasonable to report the value as halfway between those values, and the uncertainty as half the difference between the 10.5 and 10.0 mm. After rounding, that would be 10.3 mm with a standard uncertainty of 0.3 mm. If you were very confident (95% confident) that the length was between 10.0 and 10.5 mm, it would be better to report it is as 10.3 mm with a standard uncertainty of 0.1 mm. On the other hand, if you were measuring a length on a 3D object with a meter stick, issues associated with alignment and viewing angle could easily give rise to an uncertainty of several mm (more for an oddly-shaped object), despite the meter stick being marked in mm.

EXAMPLE: The half-division "rule". It is common to see uncertainty reported as half the smallest division of an analog instrument, or half the last digit of a digital instrument. While frequent, this is not actually part of the GUM standard. Uncertainty of half a division is about typical for many analog measurements, and slightly too large for most digital measurements (see above). It is usually a reasonable starting point for thinking about uncertainty. However, avoid thinking of it as a rule. You must examine the actual measurement you are performing and arrive at an honest estimate of your uncertainty.

Example: Calibration reports. Manufacturers may issue calibration report for their instruments stating the uncertainty in its calibration. It may be stated as a standard uncertainty, but more often will be stated as a 95% confidence interval $x \pm \Delta$, or sometimes a 99% confidence interval. The standard uncertainty is half of Δ for a 95% interval, or one third for a 99% interval.

Example: Systematic effects. You should always correct measurements for known systematic effects. The uncertainty reflects the uncertainty in that correction, not its magnitude. For example, let's say you're timing how long it takes for a soap bubble to pop after being created. You have no knowledge of when it will pop before it happens, but you can anticipate the bubble's creation based on when your lab partner blows on the soap film. The measured start times may therefore be early or late compared to the true start time, but the stop times will always be late. Consequently, your measured time intervals will be systematically too long. In this case, you should correct the intervals by subtracting your reaction time in your analysis. If you estimate you reaction time as 0.3 seconds with a standard uncertainty of 0.1 seconds (about typical for visual stimuli), then you should explicitly subtract 0.3 seconds from all intervals, and apply a 0.1 second associated uncertainty.

EXAMPLE: Event counting. Counting the number of random events that occur in a given time (e.g. the number of radioactive decays observed in a given time) is subject to statistical fluctuation. Some intervals will naturally have more events than others. This leads to an uncertainty in the measured event rate. The appropriate statistical distribution in this case is known as the Poisson distribution, which has a standard deviation equal to the square root of its mean. Therefore, if the observed number of counts is n then its uncertainty is $\sqrt{n}$.

Combining uncertainties

Often, there are several different sources of uncertainty u_i that contribute to an overall total uncertainty in a measurement. For example, repeated measurements might give a Type A uncertainty of u_1, calibration uncertainty (Type B) might be provided by the manufacturer as u_2, and an uncertainty in correcting for systematic effects (Type B) might be u_3. Assuming these uncertainties are uncorrelated, they can be combined into a **combined standard uncertainty** u_c by adding in quadrature:

$$u_c = \sqrt{u_1^2 + u_2^2 + u_3^2 + ...} \tag{4}$$

In practice, one uncertainty source often dominates over the others, in which case the combined uncertainty is essentially just the dominant uncertainty. For example, if you were measuring your height, Type A uncertainties associated with posture, definition of the points to measure, etc. would likely dominate.

It is important not to "double-count" your uncertainties. The final uncertainty should be an honest assessment of your doubt about the measurement result, neither unrealistically large nor unrealistically small. Most students tend to overestimate their uncertainty, incorrectly thinking of it as a sort of "safety factor". There is no merit to unrealistically inflating your uncertainty estimates.

Propagating uncertainties

Sometimes, the quantity of interest must be calculated from one or more measurements. For example, you might find the volume of a cylinder by measuring its height and diameter. The measurement uncertainty in the height and diameter must then be transferred to the

final uncertainty in the volume, a process known as **propagation** of the uncertainty.

If the final quantity r is a function of several variables $x, y, ...$ with uncertainties $u(x), u(y), ...$ then the propagated standard uncertainty u_c is

$$u_c(r) = \sqrt{\left(\frac{\partial r}{\partial x}\right)^2 u^2(x) + \left(\frac{\partial r}{\partial y}\right)^2 u^2(y) + ...} \quad (5)$$

EXAMPLE: You find the volume of a cylinder by measuring its diameter and height. If the diameter is 5.0 cm with an uncertainty of 0.1 cm, and the height is 10.03 cm with an uncertainty of 0.05 cm, then the volume is $V = (\pi/4)D^2h = 197$ cm^3 with an uncertainty

$$\begin{aligned} u_c(V) &= \sqrt{\left(\frac{\partial V}{\partial D}\right)^2 u^2(D) + \left(\frac{\partial V}{\partial h}\right)^2 u^2(h)} \\ &= \sqrt{\left(\frac{\pi Dh}{2}\right)^2 u^2(D) + \left(\frac{\pi D^2}{4}\right)^2 u^2(h)} \\ &= 8 \text{ cm}^3 \end{aligned}$$

Reporting uncertainty and significant figures

Reporting a measurement with a precision that greatly exceeds its measurement uncertainty gives a false sense of accuracy, and should be avoided. For example, reporting a length as "10.123456 meters with a standard uncertainty of 1 meter" is misleading, since all of the numbers right of the decimal place are essentially meaningless.

Unless there are good reasons to do otherwise (for example, extensive statistical analysis on a large sample size), uncertainties should be reported to one significant digit. Measurements and uncertainties should be reported in the same units, and with the same number of decimal places. Use scientific notation if appropriate. For example, the following are appropriate ways of reporting values:

- 0.8 m with a standard uncertainty of 0.2 m
- 0.0234 mm with a standard uncertainty of 0.0003 mm
- 1.234×10^8 m/s with a standard uncertainty of 0.001×10^8 m/s

Although it is very common to report uncertainties using the $\pm$ symbol (e.g. a length of $[0.4 \pm 0.1]$ mm), the GUM standard advises against this practice. The $\pm$ symbol is most often used to report an expanded uncertainty (i.e. the standard uncertainty multiplied by a coverage factor k, usually 2–3).

Comparing a measurement with an expected value

Many experiments involve comparing a measured value x to an expected value μ. In the language of statistics, such experiments "test the hypothesis that $x = \mu$" by asking "what is the likelihood that x and μ differ by at least the amount observed?" If that likelihood is very low (say, less than 5%) then the hypothesis is rejected and the values are said to disagree.

If the experiment is dominated by well-characterized Type A errors, then there is a mathematically rigorous way to test the hypothesis known as "Student's t-test"; details are covered in introductory undergraduate statistics courses. However, this is not a statistics course. Moreover, the experiments in this course will not usually be dominated by well-characterized Type A errors, which makes rigorous statistical hypothesis testing very difficult. This section covers a reasonable approximation for use in this course; it balances ease of use and general applicability against statistical rigour.

First, calculate the normalized difference

$$t = \frac{x - \mu}{u_c(x)} \tag{6}$$

where $u_c(x)$ is the uncertainty in x. A large value of t means that x and μ are very dissimilar, while a small value of t means that x is very similar to μ.

We then make the assumption that x is distributed as a normal distribution with standard deviation $u_c(x)$. Although not rigorously justified in most cases, we have constructed $u_c(x)$ specifically to make this assumption as defensible as possible, and removing this assumption requires a level of statistical depth well beyond this course.

Under the normal assumption, there is a $\sim 68\%$ chance of observing $|t| < 1$, a $\sim 95.4\%$ chance of observing $|t| < 2$ and a $\sim 99.7\%$ chance of observing $|t| < 3$. The possibility that x and μ agree simply by chance will always exist; it is up to the experimenter to decide how much they can accept that likelihood, and hence what value of $|t|$ is required before saying that x and μ "disagree". For university labs, $|t| > 2$ or $|t| > 3$ is typically taken as sufficient. Some high-stakes, high-precision experiments require much larger values before the experimenters are willing to announce a disagreement.

In this class, you can usually take $|t| > 2$ as "disagreement" between x and μ. The chance of that happening randomly is less than about 5%.

An extremely small value of $|t|$ is a signal that something likely went wrong, and you probably overestimated your uncertainties. For example, the likelihood of $|t|<0.01$ is only 0.8%, so it is quite unlikely that you should find such complete agreement.

EXAMPLE: You measure the speed of light as 2.970×10^8 m/s with a standard uncertainty of 0.005×10^8 m/s. Does your value agree with the accepted value of 2.9979×10^8 m/s?

The normalized difference is

$$t = \frac{x - \mu}{u_c(x)} = \frac{2.970 \times 10^8 - 2.9979 \times 10^8}{0.005 \times 10^8} = -5.58$$

so despite being within 1%, the measurement differs from the accepted value by more than 5 standard uncertainties. It therefore does NOT agree with the accepted value.

Comparing two measurements with uncertainties

Other experiments may require you to compare two measured values (with associated uncertainties) against each other. Again, there is a statistically rigorous way to do this in the case of Type A uncertainties (Welch's t-test), and again we will use a simplified method here.

Compute the value of t in this case as:

$$t = \frac{x_1 - x_2}{\sqrt{u_c^2(x_1) + u_c^2(x_2)}} \tag{7}$$

which reduces to equation 6 if the uncertainty in one of the measurements is taken as zero. Then, simply use the the same argumentation as above: if $|t|$ is less than about 2 or 3, the values can be said to agree.

Line fitting

Least-squares fitting

Finally, there are many experiments that seek to quantify a linear relationship between two quantities. For example, let's say you take measurements of the speed v_i of an object in freefall at different times t_i after releasing it (blue points in Figure 1). Since the acceleration is constant, the speed increases linearly with time, and it makes sense to interpret the data as having random fluctuations from a straight line $v = gt + v_0$. If this line can be determined from the data, then its slope is a measurement of the gravitational acceleration g.

Line fitting is the problem of finding a line that describes the data well, as shown in Figure 1. Intuitively, the problem is equivalent to minimizing the distances between the points and the line. The most commonly-used method is "lest-squares" fitting, where the minimized quantity is the sum of the squares of the vertical distances (blue lines

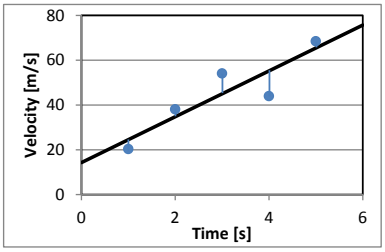


Figure 1: Measurements of the speed of an object in freefall versus time since the object's release. A least-squares line of best fit is also shown, with the equation $v = (10.2\ m/s^2)\ t + (14\ m/s)$ determined from equation 9.

in Figure 1):

$$\chi^2 = \sum_{i=1}^{n} [y_i - (mx_i + b)]^2 \tag{8}$$

for data points (x_i, y_i) fit by the line $y = mx + b$.

You could well ask if this is always the "best" definition (Why square the distances? Why the vertical distances?), but there are two compelling arguments in its favour. First, its minimization is comparatively straightforward, with the closed-form analytic solution

$$m = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sum(x_i - \bar{x})^2} \qquad\qquad b = \bar{y} - m\bar{x} \tag{9}$$

where $\bar{x}$ and $\bar{y}$ are the average values of x_i and y_i respectively and the summations are over $i = 1$ to n. Second, it's often a very defensible choice; the solution in equation 9 turns out to be the most probable line if the errors in y are normally-distributed and the errors in x are negligible[1].

Least-squares fitting using equation 9 (or similar) is implemented in many pieces of software. In Excel, use the LINEST function, taking care that the formula is a bit tricky to enter since LINEST returns multiple values (it's an "array formula"). There is a 1-page LINEST help sheet available on D2L.

Nonlinear relationships

Although many experimental relations can be fit with a function of the form $y = mx + b$ ("linear relations"), there are also plenty that cannot. For example, you might measure the position y as a function of time t for an accelerated mass, in which case the expected behaviour is $y = (a/2)t^2$, which is not linear in y and t and cannot immediately be fit as $y = mx + b$.

You should always examine a plot of your data to check whether it makes sense to fit a line to it. Does it actually resemble a line (plus random deviations)? Random scatter about the line like in Figure 1 is normal. A few points far from the line (outliers) may indicate a minor problem. Systematic curves or discontinuities indicate a serious problem.

You may be able to linearize the relation by a slight rearrangement. In the above example, while a plot of y versus t would not be a straight line, a plot of y versus t^2 would. Therefore, you could simply fit the line $y = mx + b$ using $x = t^2$. The resulting slope m would be $(a/2)$ and the resulting offset b would be the initial starting position.

Sometimes a simple algebraic rearrangement is required. For example, power-law relations like $y = Ax^B$ can be fit by taking the logarithm of both sides of the equation to yield $\ln y = \ln A + B \ln x$. Fitting a linear line $y' = mx' + b$ with $y' = \ln y$ and $x' = \ln x$ then gives both $\ln A$ and B.

Not all nonlinear functions can be treated in this way. If we continue our accelerated-mass example but allow an initial velocity v_0, then the expected behaviour becomes $y = v_0 t + (a/2)t^2$, which cannot be

rearranged into the form $y = mx + b$. In such a case, we would need curve-fitting methods not covered in this document.

Type-A uncertainty in the fitted coefficients

There is clearly an uncertainty associated with the values of the fitted coefficients m and b. Figure 2 shows several lines, each with a different slope, any of which could plausibly fit the data. The solid line may be the most probable, but the dashed lines are still plausible.

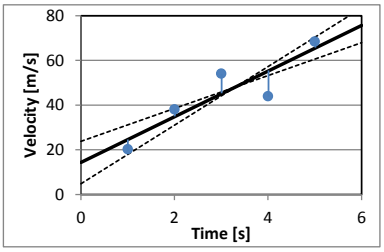


Figure 2: same as Figure 1 with the addition of two dashed lines. The dashed lines have slopes equal to the fitted slope $m = 10.2$ plus or minus the standard error in the slope $u(m) = 2.9$ in order to show the range of possible lines of best fit.

Recall that Type-A uncertainty is derived from the scatter of repeated measurements about a mean value, and is equal to the standard deviation (eq. 3). In exactly the same way, we can define a Type-A uncertainty in m based on the scatter of repeated measurements about a line. Intuitively, this makes sense: we should become less certain about m as the scatter of the points about the line increases. This Type-A uncertainty is given by

$$u(m) = \frac{\sum[y_i - (mx_i + b)]^2}{(n-2)\sum(x_i - \bar{x})^2} \qquad u(b) = u(m)\sqrt{\frac{1}{n}\sum x_i^2} \tag{10}$$

where the summations are over $i = 1$ to the number of data points n.

In the literature, the Type-A uncertainty estimate in equation 10 is usually (and perhaps confusingly) called the "standard error" for the coefficients. It is implemented as part of Excel's LINEST function.

Line fitting with measurement uncertainty

The uncertainty estimate in the previous section has a few important restrictions:

1. It is strictly Type-A, and cannot account for fundamentally Type-B uncertainties such as uncertainties in calibration or uncertainties in correcting for a systematic effect.

2. It weights all points equally, assuming they all have the same uncertainty.

3. It assumes the x_i values are exact, attributing all uncertainty to y_i.

These restrictions may or may not be acceptable, depending on the experiment. Note that these restrictions potentially affect not only the uncertainties but also the values of m and b.

If the above restrictions are not acceptable, then your experiment calls for a more sophisticated uncertainty estimate. In this case, the first two restrictions in the list can be lifted by propagating uncertainty values through the linear line-fitting equations. This allows for the

inclusion of Type-B uncertainties, and allows for unequal uncertainties for the individual points. The equations that result from this error propagation are listed in Appendix A. (Don't worry, you don't need to derive them yourself.)

The equations in Appendix A are implemented in the Excel file "Line fitting template.xlsx" on D2L. Feel free to use this as a template for any line-fitting required in this course.

The last restriction in the list (assuming the x_i values are exact) comes from using the vertical distances (e.g. Figure 1 or 2) instead of the uncertainty-weighted shortest distances in the definition of χ^2 (equation 8). It becomes a valid assumption only if

$$u(y) \gg \frac{dy}{dx} u(x) \tag{11}$$

which is only sometimes true. If the converse is true (i.e. $u(x)$ dominates), consider simply reversing x and y. If the uncertainties are comparable, then you may combine the uncertainty in $u(x)$ with the uncertainty in $u(y)$:

$$\sigma_i = \sqrt{u^2(y_i) + \left[\frac{dy}{dx} u(x_i)\right]^2} \tag{12}$$

and then use σ in place of $u(y)$.

EXAMPLE: count rate as a function of distance. Reference [1] reports data from an experiment in which a radioactive source was moved away from a detector in 10 steps. At each step, the distance r between source and detector and the number of counts n detected in 450 seconds was recorded.

The raw data accumulated was:

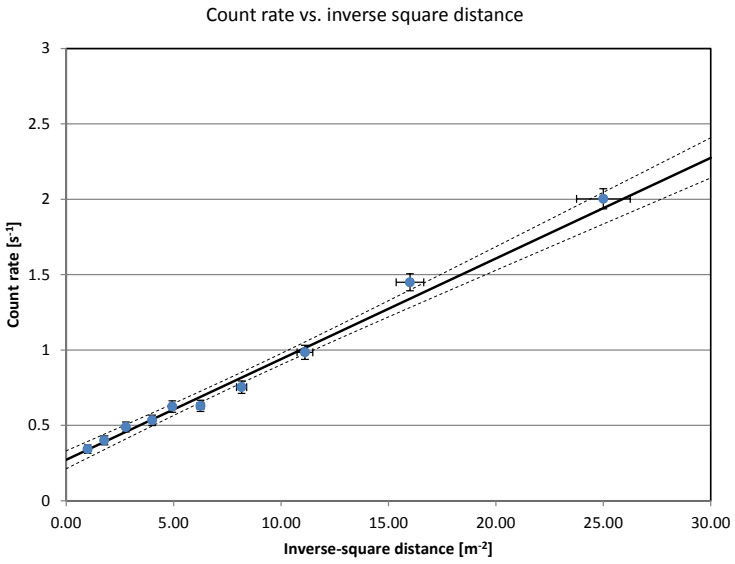
Distance r_i [m]	Counts n_i per 450 s [unitless]
0.200	901
0.250	652
0.300	443
0.350	339
0.400	283
0.450	281
0.500	240
0.600	220
0.750	180
1.000	154

The uncertainty in the distance $u(r_i)$ should be estimated with a Type B uncertainty by examining the apparatus. We will assume it to be 0.5 cm for this example. The uncertainty in the number of counts should be calculated as $u(n_i) = \sqrt{n_i}$ (see example on page 7).

EXAMPLE (continued). Theoretically, the measured count rate should consist of a constant background plus counts from the source that decrease as $1/r^2$. To fit this with a linear function, we will therefore fit count rate versus $1/r^2$. The propagated values and uncertainties are shown in the table below. The inverse-square distance $x_i = 1/r_i^2$ is calculated from r_i, and its associated uncertainty is propagated using equation 5 as $u(x_i) = 2u(r_i)/r_i^3$. Count rates are computed by dividing n_i by 450 seconds (and similarly for $u(n_i)$).

distance [m]		inverse-square distance [m^{-2}]		Count rate [s^{-1}]	
r_i	$u(r_i)$	$x_i = 1/r_i^2$	$u(x_i)$	y_i	$u(y_i)$
0.200	0.005	25.00	1.25	2.00	0.07
0.250	0.005	16.00	0.64	1.45	0.06
0.300	0.005	11.11	0.37	0.98	0.05
0.350	0.005	8.16	0.23	0.75	0.04
0.400	0.005	6.25	0.16	0.63	0.04
0.450	0.005	4.94	0.11	0.62	0.04
0.500	0.005	4.00	0.08	0.53	0.03
0.600	0.005	2.78	0.05	0.49	0.03
0.750	0.005	1.78	0.02	0.40	0.03
1.000	0.005	1.00	0.01	0.34	0.03

The uncertainties (both $u(x_i)$ and $u(y_i)$) vary substantially from point to point, and the uncertainty in x is considerable. The line of best fit should therefore be calculated using the equations in Appendix A, which results in the line of best fit shown below. Detailed calculations are shown in the Excel document "Linefitting_example.xlsx" on D2L. The fitted slope is 0.067 m^2/s with a standard uncertainty of 0.003 m^2/s, and the fitted intercept is 0.27 s^{-1} with a standard uncertainty of 0.02 s^{-1}. The dashed curves show the range of lines consistent with these uncertainties, using a coverage factor $k = 2$ (95% confidence) (see Appendix A).



Appendix A: Line fitting equations

If the uncertainty in x $u(x)$ is significant (as assessed by equation 11), then the combined uncertainty σ_i should be computed using equation 12. Otherwise, let $\sigma_i = u(y_i)$. Then, compute

$$\Delta = \sum \frac{1}{\sigma_i^2} \sum \frac{x_i^2}{\sigma_i^2} - \left(\sum \frac{x_i}{\sigma_i^2} \right)^2 . \tag{13}$$

with the summations over $i = 1$ to n. Next, compute m and b as

$$m = \frac{1}{\Delta} \left(\sum \frac{1}{\sigma_i^2} \sum \frac{x_i y_i}{\sigma_i^2} - \sum \frac{x_i}{\sigma_i^2} \sum \frac{y_i}{\sigma_i^2} \right) \tag{14}$$

$$b = \frac{1}{\Delta} \left(\sum \frac{x_i^2}{\sigma_i^2} \sum \frac{y_i}{\sigma_i^2} - \sum \frac{x_i}{\sigma_i^2} \sum \frac{x_i y_i}{\sigma_i^2} \right) \tag{15}$$

and finally the uncertainties as

$$u(m) = \sqrt{\frac{1}{\Delta} \sum \frac{1}{\sigma_i^2}} \tag{16}$$

$$u(b) = \sqrt{\frac{1}{\Delta} \sum \frac{x_i^2}{\sigma_i^2}} \tag{17}$$

For lines fitted in this manner, data can be expected to lie within the range

$$y = mx + b \pm k \sqrt{[u(b)]^2 + [(x - \bar{x}_i) u(m)]^2} \tag{18}$$

roughly 68% of the time for a coverage factor k=1 and 95% for $k = 2$.

For further details on these equations, see Bevington and Robinson[1], chapter 6. The equations are also implemented in the Excel spreadsheet available on D2L.

References

[1] P.R. Bevington and D.K. Robinson. *Data Reduction and Error Analysis for the Physical Sciences*. McGraw-Hill, 2nd edition, 1992.

[2] Joint Committee for Guides in Metrology. JCGM 100: Evaluation of measurement data - guide to the expression of uncertainty in measurement. Technical report, JCGM, 2008. [www.bipm.org/en/publications/guides/gum.html, also available on D2L].